# Customer Segmentation

Credit Card User Behavior Analysis

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ML II

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# Executive Summary

This report presents the findings of a comprehensive customer segmentation analysis performed on credit card user data. Using advanced machine learning clustering techniques, we have identified seven distinct customer segments based on their spending patterns, payment behaviors, and credit utilization characteristics.

Each segment represents a unique behavioral profile with specific risk levels, revenue potential, and strategic implications. This segmentation enables targeted marketing strategies, risk management approaches, and personalized product offerings to maximize customer lifetime value while minimizing default risk.

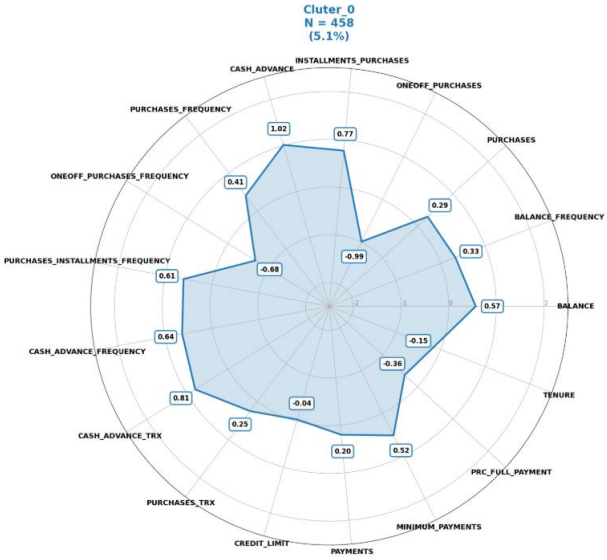
### Key Findings at a Glance

| Cluster | Segment Name | Risk Level | Revenue Potential |
|---|---|---|---|
| 0 | Structured Installment Users | Medium | Good |
| 1 | Responsible Installment Planners | Low | Moderate |
| 2 | High-Utilization Credit Reliant | High | Very High |
| 3 | Cash-Advance Dependent | Very High | Moderate |
| 4 | Premium Power Users | Medium | Very High |
| 5 | Retail-Focused Transactional | Low-Medium | High |
| 6 | Occasional Big-Ticket Spenders | Low-Medium | Good |

# Cluster Profiles

## Cluster 0: Structured Installment Users

● Medium Risk | ● High Revenue Potential




### Observed Patterns

- Moderate BALANCE
- Medium-High INSTALLMENTS_PURCHASES
- Moderate PURCHASES
- Low-Moderate ONEOFF_PURCHASES
- Medium CASH_ADVANCE & related frequencies
- Medium PURCHASES_TRX
- Medium PAYMENTS
- Medium PRCFULLPAYMENT
- Average CREDIT_LIMIT
- Medium tenure

### Behavioral Interpretation

This group uses the card regularly and prefers installment-based purchases. They use cash advance occasionally and make consistent payments. They are not extreme spenders or extreme risk. They show controlled but active usage behavior and are financially engaged but not aggressive.

### Revenue Potential

Good long-term value through interest revenue from installments, moderate transaction fees, and stable behavior.

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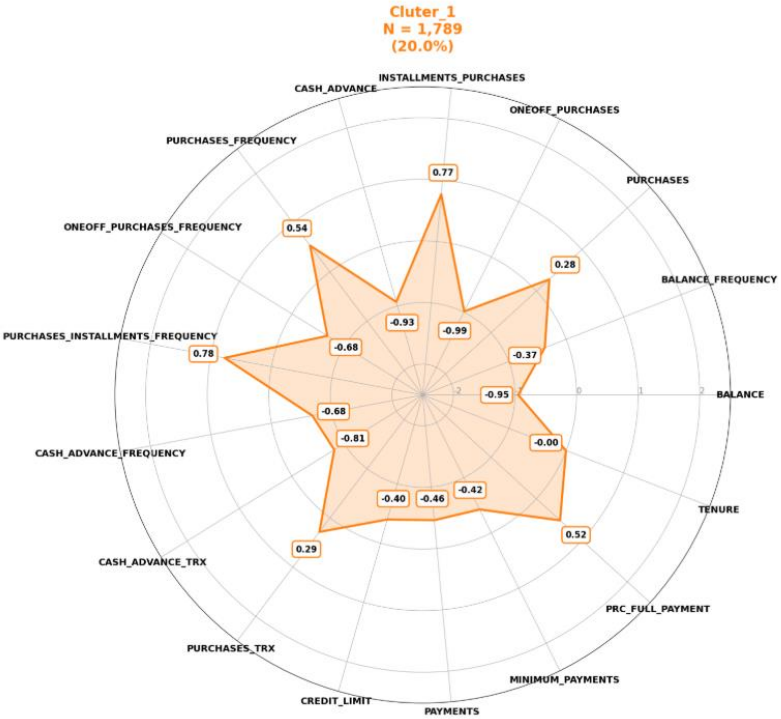
## Cluster 1: Responsible Installment Planners

● **Low Risk** | ● **Moderate Revenue Potential**




### Observed Patterns

- Lower BALANCE
- High INSTALLMENTS_PURCHASES
- Moderate PURCHASES
- Lower CASH_ADVANCE

- Lower CASH_ADVANCE_FREQUENCY
- Higher PRCFULLPAYMENT
- Medium PAYMENTS
- Lower MINIMUM_PAYMENTS
- Moderate tenure

### Behavioral Interpretation

This group uses installments heavily, avoids cash advances, and pays more responsibly with higher percent full payment. They maintain lower revolving balances. They are disciplined users who use credit as a tool, not as debt accumulation.

### Revenue Potential

Moderate revenue potential with less interest income but more transaction-based revenue and lower default risk. These are high-quality customers.

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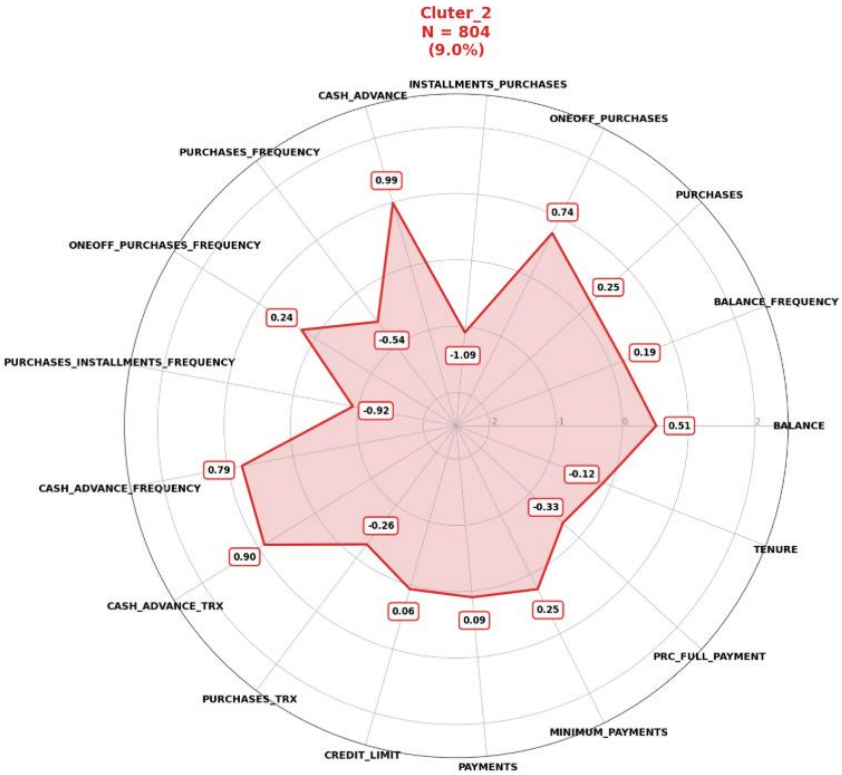
## Cluster 2: High-Utilization Credit Reliant Users

● **High Risk** | ● **High Revenue Potential**



### Observed Patterns

- Higher BALANCE
- Higher PURCHASES
- High ONEOFF_PURCHASES
- High CASH_ADVANCE
- High CASH_ADVANCE_TRX
- Medium-High PURCHASES_TRX
- Medium PRCFULLPAYMENT
- Higher MINIMUM_PAYMENTS
- Medium tenure

### Behavioral Interpretation

This cluster spends aggressively, uses one-off large purchases, and uses cash advance frequently. They carry significant balances and likely revolve debt. This group relies heavily on credit liquidity and behaves more like credit-dependent customers.

### Revenue Potential

Very high short-term revenue from interest income, cash advance fees, and transaction volume. However, risk-adjusted return may be lower.

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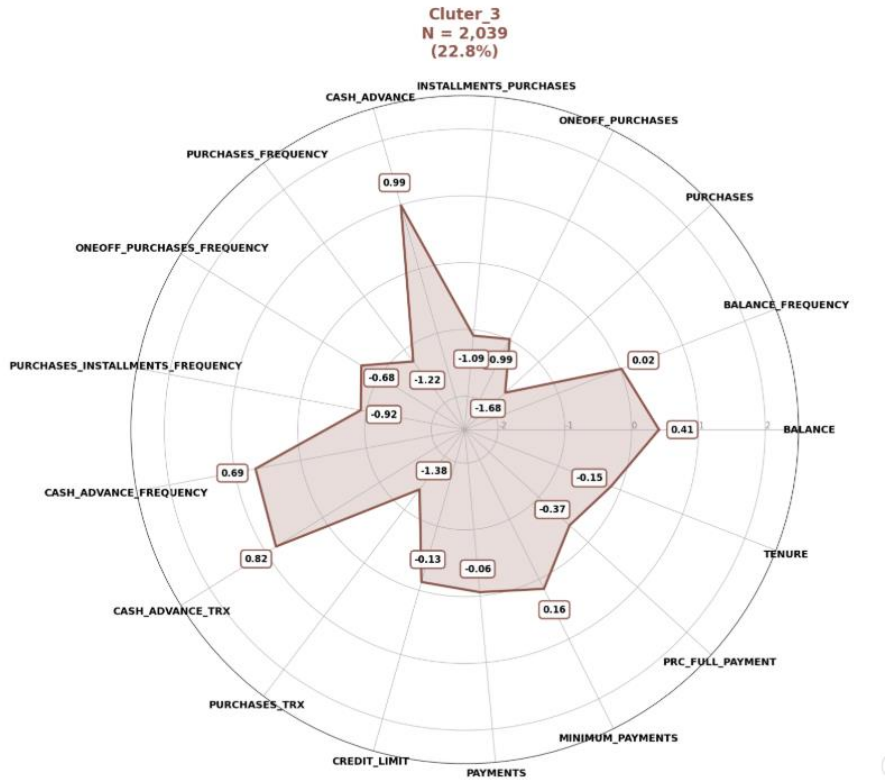
## Cluster 3: Cash-Advance Dependent Users

● **High Risk** | ● **Moderate Revenue Potential**




### Observed Patterns

- Moderate BALANCE
- Very high CASH_ADVANCE
- High CASH_ADVANCE_TRX
- High CASH_ADVANCE_FREQUENCY
- Low PURCHASES
- Low INSTALLMENTS_PURCHASES
- Low ONEOFF_PURCHASES
- Moderate PAYMENTS
- Moderate PRCFULLPAYMENT
- Moderate credit limit

### Behavioral Interpretation

This cluster uses the card primarily for cash withdrawal and is not purchase-oriented. They rely on credit as short-term liquidity and are likely financially stressed. They revolve balance. This is a cash-dependent user group, not a retail spender group.

### Revenue Potential

High short-term revenue from fees and interest, but risk-adjusted profitability may be poor. Cash advance usage is strongly correlated with default probability in credit scoring literature.

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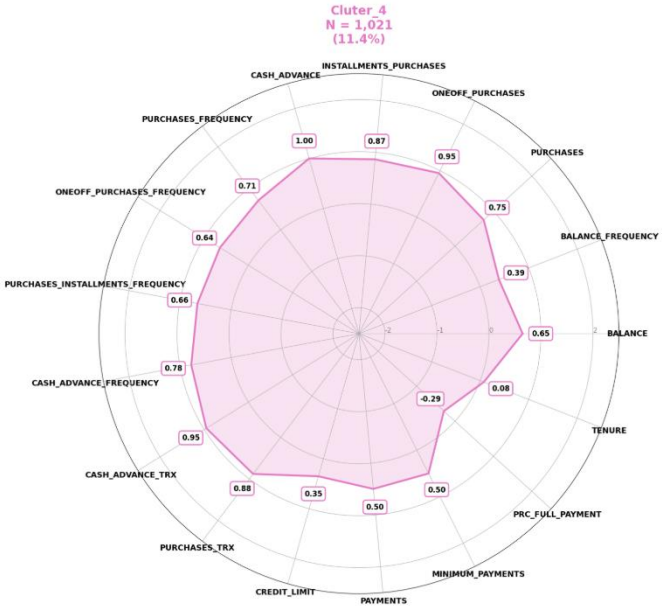
## Cluster 4: Premium High-Activity Power Users

● Medium Risk | ● High Revenue Potential




### Observed Patterns

- High BALANCE
- High PURCHASES
- High ONEOFF_PURCHASES
- High INSTALLMENTS_PURCHASES
- High CASH_ADVANCE
- High transaction frequencies

- High PAYMENTS
- High CREDIT_LIMIT
- Moderate PRCFULLPAYMENT
- Stable tenure

### Behavioral Interpretation

This is a heavy active user group that spends a lot, uses all credit features, makes large payments, has high credit limits, and is engaged in all transaction types. They are power users.

### Revenue Potential

Very high revenue potential through high interchange revenue, high interest potential, and high engagement. These are very valuable customers.

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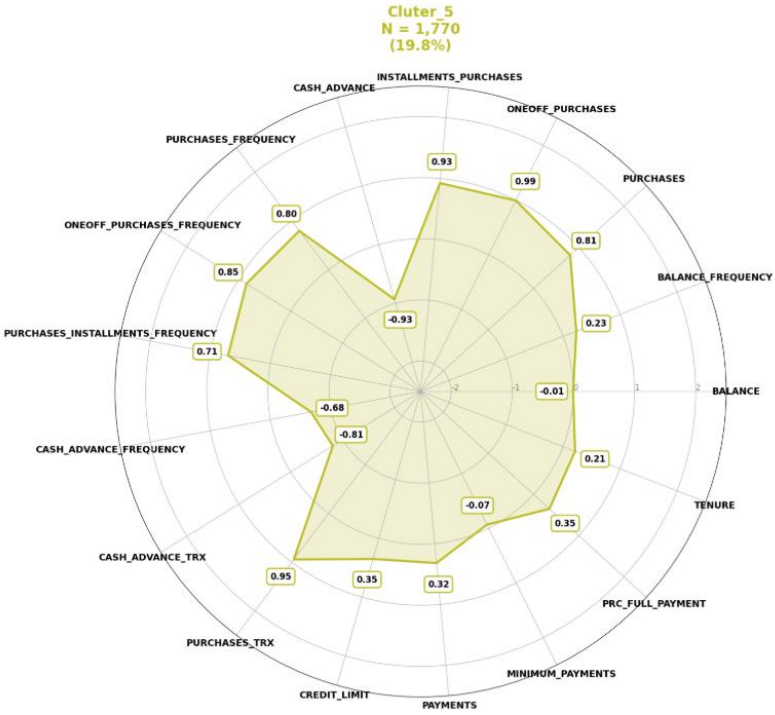
## Cluster 5: Retail-Focused Transactional Users

● Medium Risk | ● High Revenue Potential




### Observed Patterns

- Moderate-High PURCHASES

- Moderate BALANCE
- Low CASH_ADVANCE_FREQUENCY
- Low-Moderate CASH_ADVANCE
- Higher PURCHASES_TRX
- Moderate INSTALLMENTS_PURCHASES
- Medium PRCFULLPAYMENT
- Medium credit limit
- Balanced tenure

### Behavioral Interpretation

This group is purchase-focused, not cash-advance focused, transaction-heavy, and likely everyday shoppers with balanced repayment behavior. They are retail active users, not debt-heavy users.

### Revenue Potential

High revenue potential via transaction fees and moderate interest revenue.

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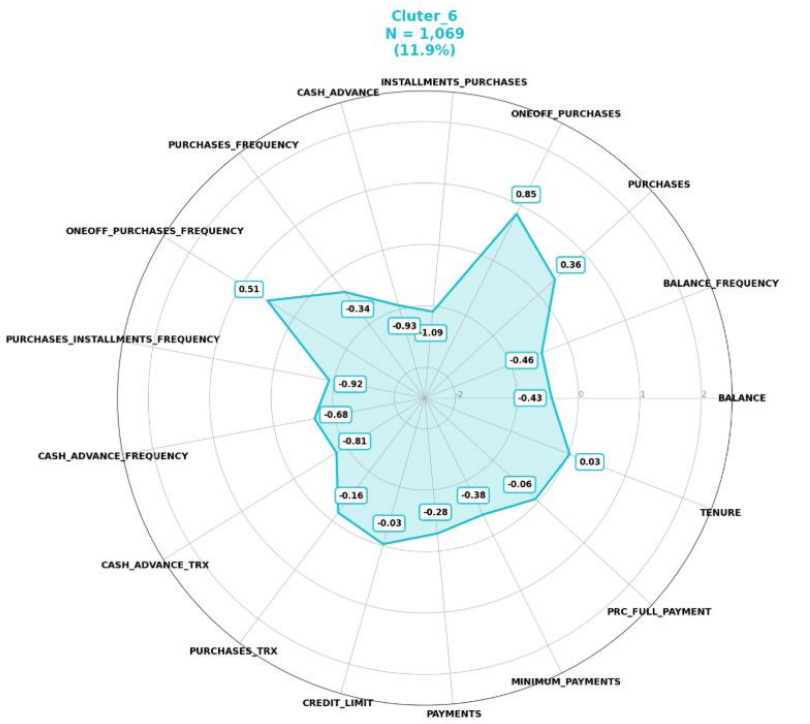
## Cluster 6: Occasional Big-Ticket Spenders

● Medium Risk | ● High Revenue Potential




### Observed Patterns

- Moderate PURCHASES
- Moderate-High ONEOFF_PURCHASES
- Low-Moderate INSTALLMENTS_PURCHASES
- Low CASH_ADVANCE
- Low CASH_ADVANCE_FREQUENCY
- Low CASH_ADVANCE_TRX
- Moderate PURCHASES_FREQUENCY
- Moderate PURCHASES_TRX
- Moderate BALANCE
- Moderate PAYMENTS
- Moderate PRCFULLPAYMENT
- Medium credit limit

- Average tenure

### Behavioral Interpretation

This group uses the card mainly for discretionary or occasional large purchases, does not rely on installments heavily, avoids cash advances, maintains manageable balances, and shows stable repayment behavior. They are controlled spenders, possibly middle-income professionals using credit for convenience rather than financing needs. They are not liquidity-dependent.

### Revenue Potential

Good transaction-based revenue and moderate interest revenue. Revenue mainly from one-off purchases, merchant interchange, and occasional interest.

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